Where the Gigabytes Went · AS

U02b · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Questions labelled Ours are authored. Show working and attach units. Unless stated otherwise, count uncompressed pixel/sample data only, excluding headers, metadata and compression. Sound recordings have one channel.

1. Ours: choose the unit

(a) State the number of bits in a byte and in a nibble. **[2]**

(b) Convert 1 kB to bits. **[1]**

(c) Convert 1 KiB to bits. **[1]**

2. Ours: a scanned bitmap

(a) A bitmap is 1024 × 768 pixels at 24 bits per pixel. Estimate its uncompressed pixel-data size in MiB. **[3]**

(b) State one item a bitmap header can describe. **[1]**

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3. Ours: a sound clip

(a) A 32-second clip uses 16 384 samples per second and 16 bits per sample. Calculate its raw size in KiB. **[3]**

(b) Only the sample rate is halved. State the new raw size in KiB. **[1]**

4. Ours: find the error

> An 800 × 600 image uses 8 bits per pixel.
> A student writes: 800 × 600 × 8 ÷ 1024 = 3750 KiB.

Identify the error and give the correct size in KiB, with corrected working. **[3]**

5. Ours: two prefixes

State the number of bytes in 1 GB and in 1 GiB. **[1]**

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6. Ours: exam-shape rehearsal · prefixes

Fresh authored unit conversions, modelled on the prefix-and-value question shape.

(a) A mebibyte has a _____ prefix. Five mebibytes = _____ bytes. **[2]**

(b) A gigabyte has a _____ prefix. Seven terabytes = _____ gigabytes. **[2]**

7. Ours: compare before choosing

Which is larger: 3100 kB or 3 MiB? Show both quantities in bytes. **[2]**

8. Ours: payload is not the whole file

A file has 512 KiB of uncompressed pixels and a 1024-byte header. There is no other overhead or compression. Calculate the total file size in KiB, showing working. **[2]**

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9. Ours: predict the file-size machine · pen first [6]

💻 **Write your predictions before running the deck's file-size cell.** Start with width = 512, height = 512 and depth = 16 bits per pixel. The cell computes bits = width × height × depth, then divides by 8 and 1024 to print KiB. These six marks are human marked; no separate Python file is needed.

(a) Calculate the initial output in KiB. Show the conversion from bits. **[2]**

(b) Change only width to 1024. Predict the new output in KiB and explain its relationship to the initial output. **[2]**

(c) Restore width to 512 and change only depth to 8. Predict the output in KiB and explain why. **[2]**

Then run and compare: use the deck cell to check all three predictions. Continue **Karel 07 PotholePatrol and 08 HurdleJumper** as your keyboard companion, outside the 30 marks.

Above and beyond · unmarked. For an image you can inspect, record its dimensions, format and actual file size. Assuming 24 bits per pixel, calculate its raw-pixel estimate. Suggest why the two sizes differ; state what you cannot infer from dimensions alone.

Additional space - label the question number.